Amazon Product Analytics Report

Business Intelligence & Product Performance Analysis

This report analyzes Amazon product catalog data to identify patterns in customer engagement, pricing strategy, discount behavior, and customer satisfaction. The goal is to generate actionable insights that support business stakeholders such as category managers, pricing strategists, and product quality teams.

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Executive Summary

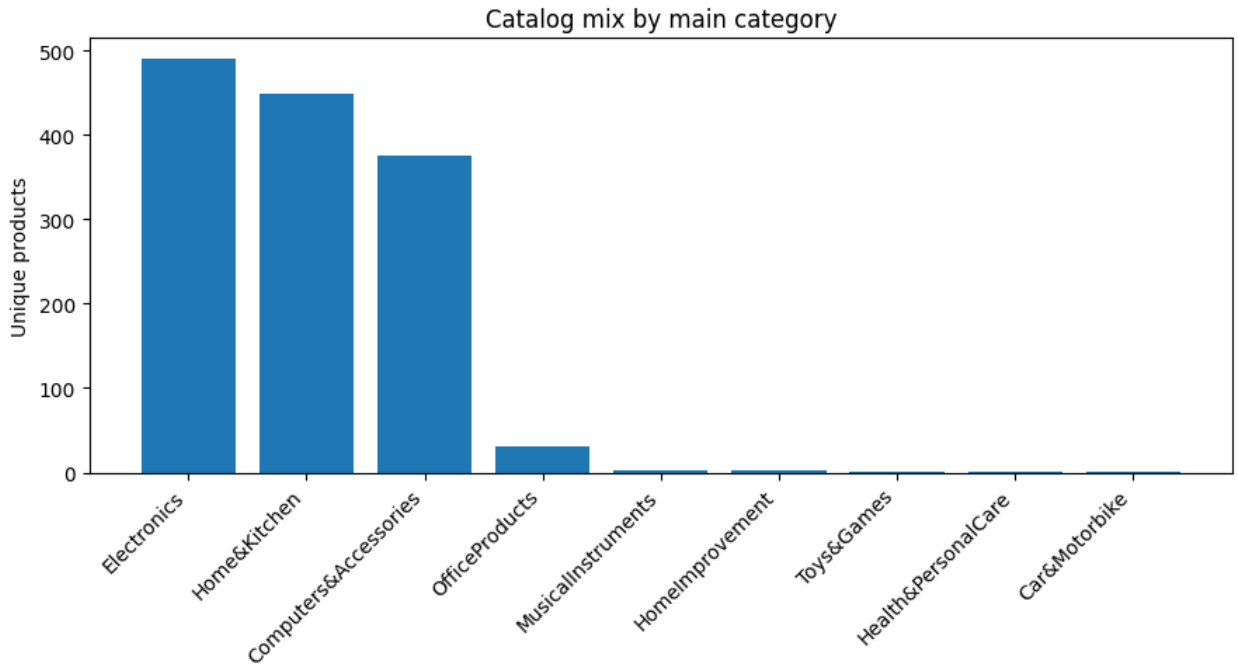
The dataset reveals strong customer satisfaction across the Amazon catalog, with ratings clustered around 4.0–4.3. Product attention is highly concentrated in a few categories (notably Electronics). Discounting strategies show weak relationships with both ratings and engagement, suggesting that heavy discounting alone does not drive customer satisfaction. Both low-price and premium products can achieve strong engagement depending on perceived value.

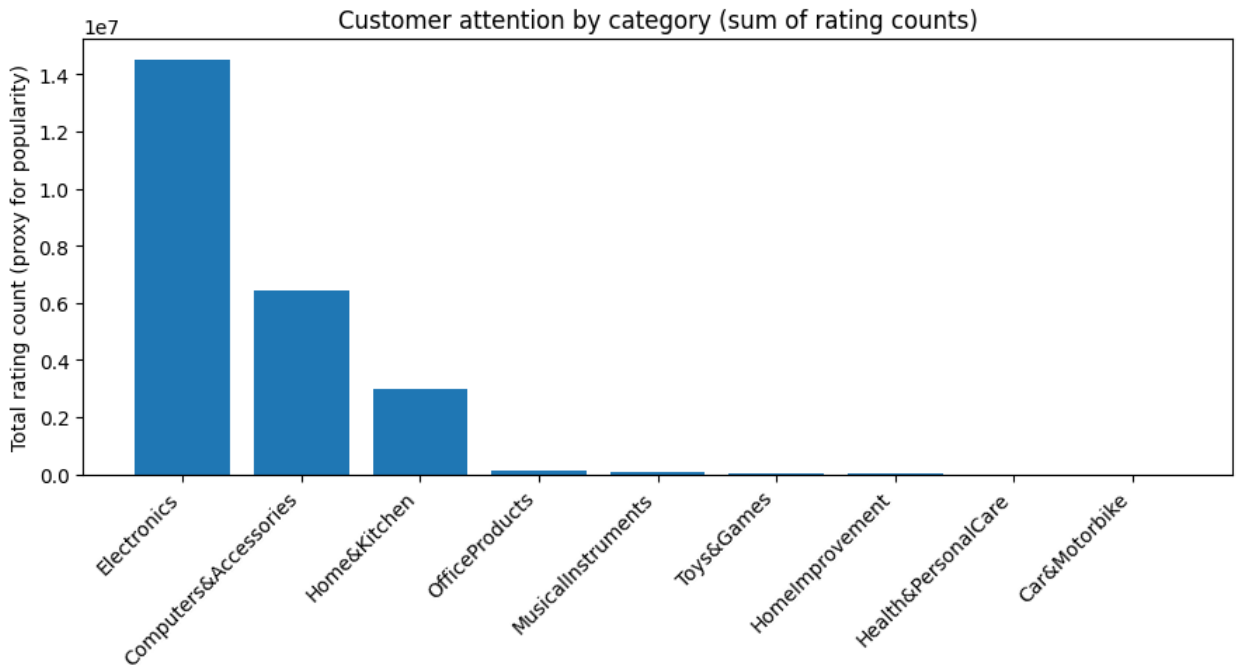
Key Insights

- Electronics and Computers & Accessories dominate customer attention.
- Customer satisfaction is generally high across products.
- Large discounts do not consistently improve ratings.
- Premium products can still achieve strong popularity.
- Product teams should focus on high visibility low rating items to improve quality.

1) Where is the catalog concentrated, and where is customer attention concentrated?

Answer: Most products are in **Electronics, Home&Kitchen, and Computers&Accessories**, and **Electronics dominates attention** (highest total rating-count, used here as a "popularity proxy").



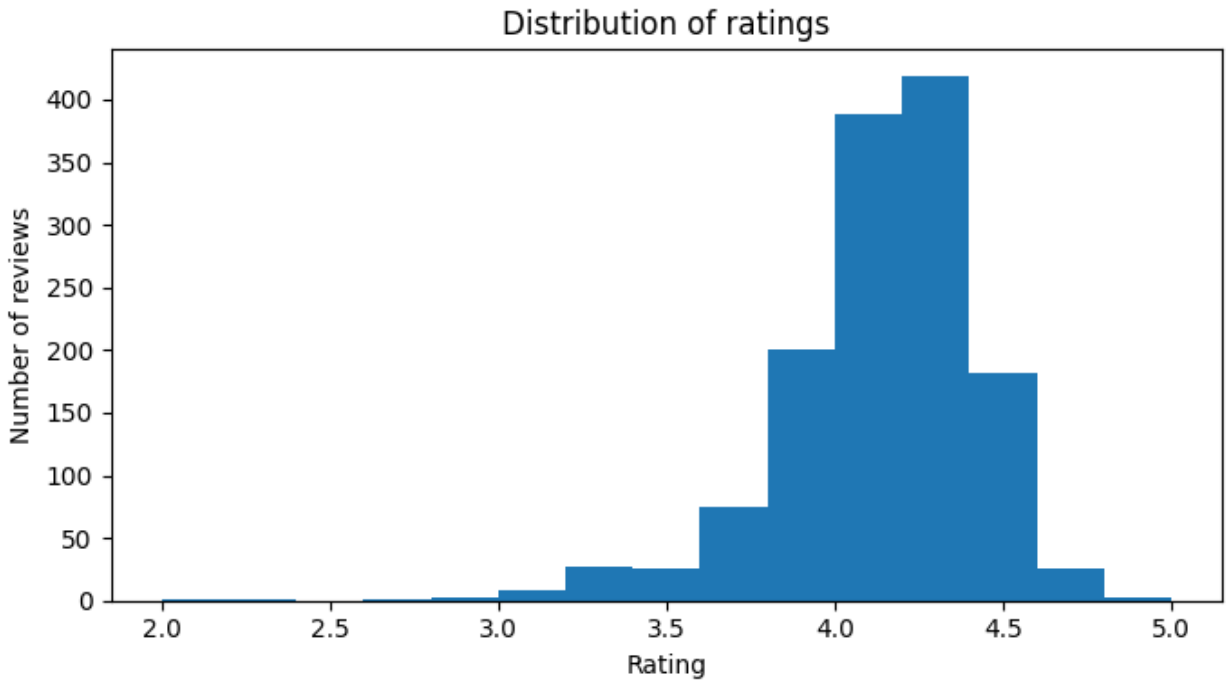


(Shown in charts: **Catalog mix by category**, **Customer attention by category**)

Stakeholder use: focus merchandising, ads, and inventory attention on categories with the biggest "attention density".

2) What does customer satisfaction look like overall?

Answer: Ratings are heavily clustered around **~4.0–4.3**, with relatively few low-rated items.



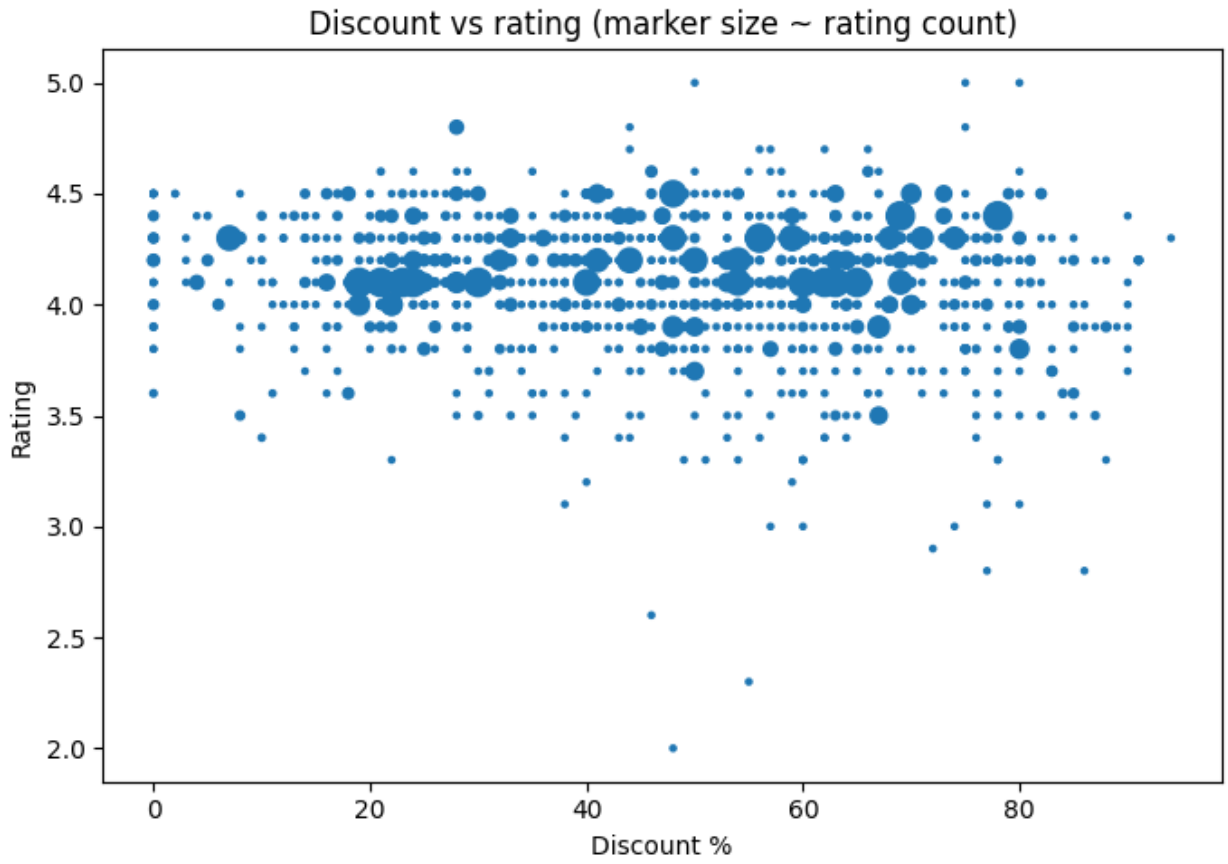
(Shown in chart: **Distribution of ratings**)

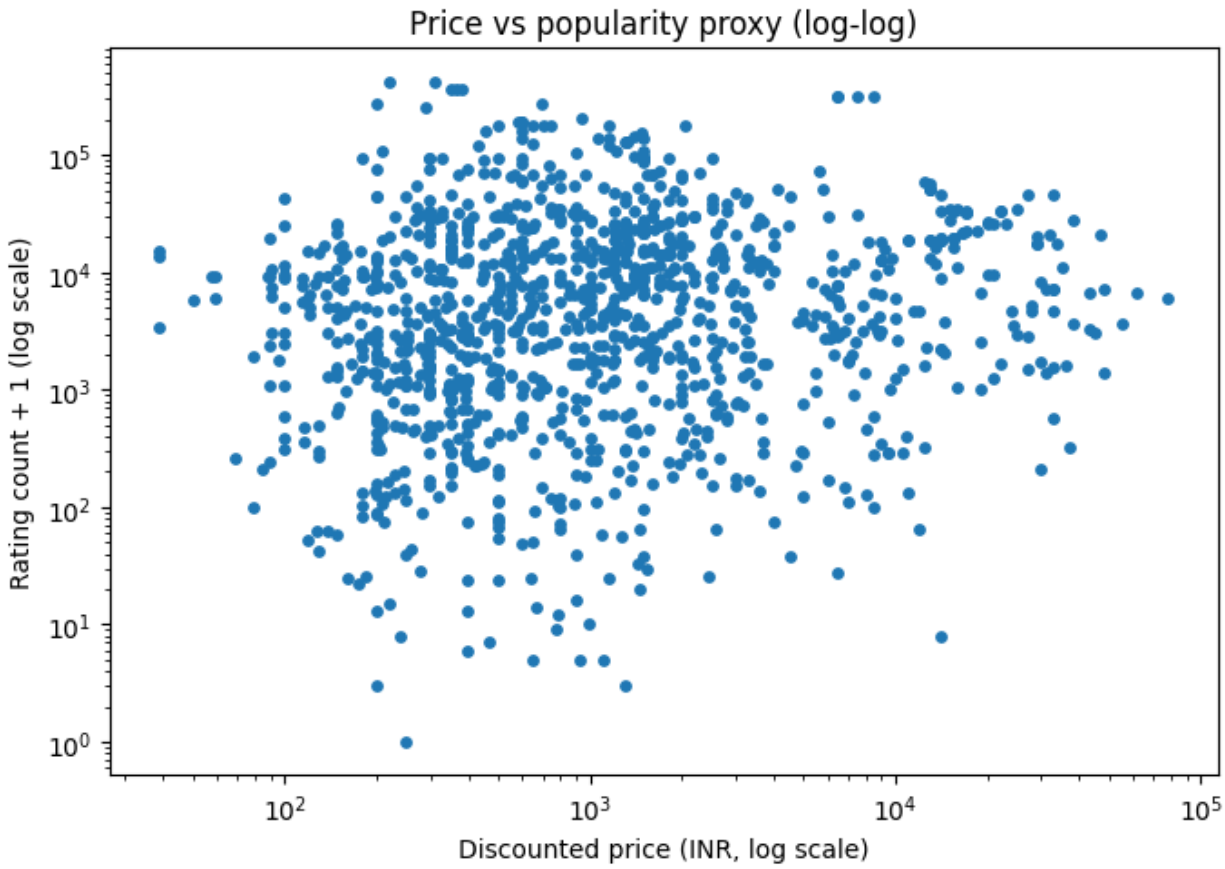
Stakeholder use: most products have decent satisfaction; isolate exceptions (see "risks" below).

3) Do bigger discounts improve ratings or popularity?

Answer (from correlation + scatter):

- Discount % has a **weak negative relationship** with rating and rating-count (popularity proxy).
- Deep discounts don't reliably mean better satisfaction; often they may be used to move weaker items.


Discount vs rating (marker size ~ rating count)
Rating
Discount %
5.0
4.5
4.0
3.5
3.0
2.5
2.0
0
20
40
60
80

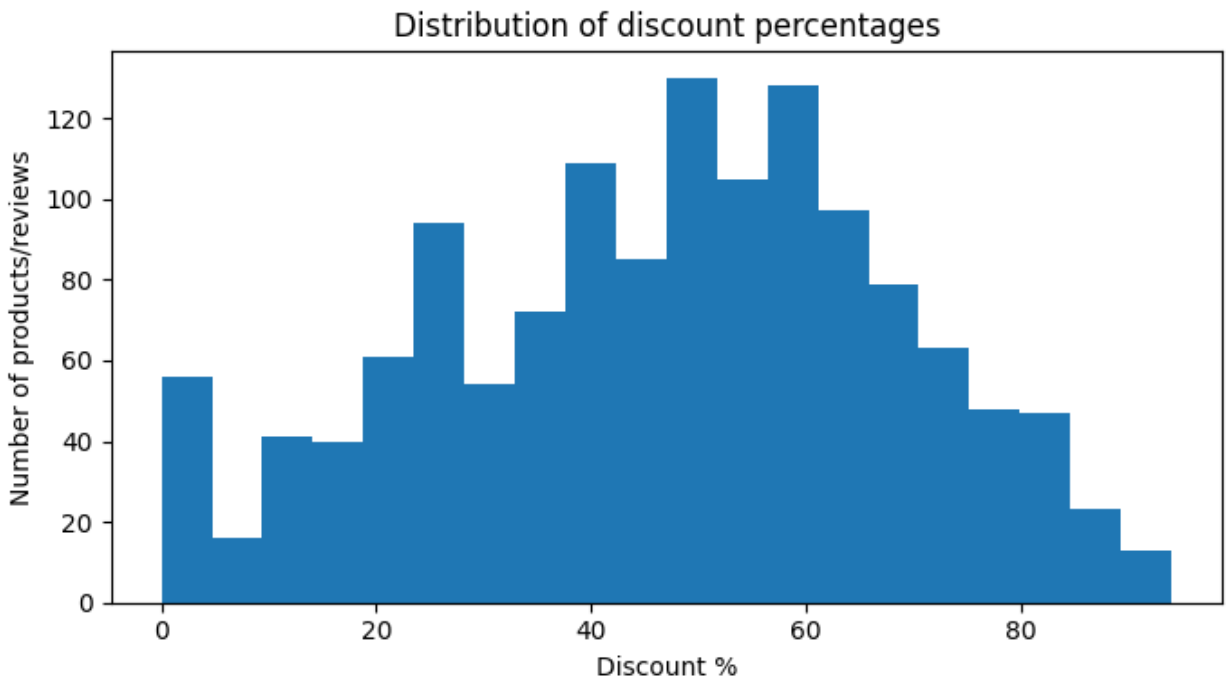


(Shown in chart: **Discount vs rating** + correlation output + **discount distribution**)

Stakeholder use: don't assume "more discount = better performance." Validate discount strategy by category and product group.

4) Does price relate to popularity?

Answer: On a log-log view, popularity spans all price points, but there's a **slight positive relationship** between price and rating-count (very weak).



(Shown in chart: **Price vs popularity proxy (log-log)**)

Stakeholder use: premium items can still be strong performers—use this to guide positioning, bundles, and placements.

5) Which products are "stars" vs "risks" (high attention but low satisfaction)?

I identified:

- **Stars:** high rating (≥ 4.3) + top 10% rating-count
- **Risks:** rating (≤ 3.8) + top 10% rating-count

These are the *most actionable* for stakeholders:

- **Stars:** push ads, bundles, cross-sell, "featured" placement
- **Risks:** investigate quality/expectation gap, listing accuracy, returns, supplier issues

Strategic Recommendations

1. Prioritize high attention categories such as Electronics for marketing investment.
2. Monitor low rating products within high visibility categories to improve quality.
3. Use data driven pricing strategies rather than relying heavily on discounting.
4. Highlight premium products with strong value propositions through improved product listings.
5. Develop dashboards to continuously track rating trends, pricing effectiveness, and category performance.

This analysis demonstrates how product catalog data can support strategic decision-making in pricing, merchandising, and customer experience optimization.